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RESEARCH ARTICLE

Economic and Technical Evaluation of Hydrogen Storage in Hybrid Renewable Systems With Demand-Side Management: Upper Egypt Case Study

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ABSTRACT To address the intermittent nature of solar photovoltaic (PV) and wind energy systems, the deployment of multiple energy storage facilities has been significantly expanded, enhancing power system reliability and flexibility toward sustainable energy solutions. This paper focuses on analyzing energy systems that utilize different energy storage options, including battery energy storage system (BESS) and hydrogen energy storage (HES). In this context, this study aims to evaluate the techno-economic and environmental impacts of integrating a hydrogen energy storage (HES) facility comprising an electrolyzer, fuel cell, and hydrogen tank into a hybrid PV/wind/battery energy storage system (BESS). Three different systems have been considered in this analysis. In this paper, three systems are considered: a diesel generator, a hybrid renewable energy system without demand-side management (DSM), and a hybrid renewable energy system with DSM. This study aims to develop a model for electricity generation based on various combinations of hybrid renewable energy systems (HRES) using HOMER energy software in Upper Egypt, specifically focusing on Qena City and Hurghada City. The primary objectives are to identify the optimal configuration of a hybrid renewable energy system that reliably, continuously, and sustainably meets the load requirements of these locations, while minimizing the total system's net present cost and the cost of energy (COE) through multi-objective analysis with HOMER Pro software. The simulation results using HOMER Pro software indicate that adopting demand-side management (DSM) results in the lowest net present cost (NPC) values for both Qena and Hurghada, at \$798,614 and \$646,046, respectively. Additionally, hydrogen production increases to approximately 953 kg/year in Qena and 850 kg/year in Hurghada, alongside reductions in post-DSM load and carbon dioxide (CO_2) emissions. The comparative analysis shows that the diesel system contributes 124,654 kg/year of CO_2 emissions. In contrast, the proposed hybrid system with DSM significantly reduces CO_2 emissions to 0.275 kg/year in Qena and 0.0666 kg/year in Hurghada.

INDEX TERMS DSM, HOMER, hybrid energy system, hydrogen system, PV, wind.

I. INTRODUCTION

Recently, numerous challenges in electrical energy production have been observed globally, particularly in rural areas

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where diesel generators are commonly used to meet energy demands, leading to increased carbon emissions. Consequently, it is crucial to integrate renewable and sustainable energy sources, such as solar and wind energy, into utility grids. This shift aims to expand the share of renewable energy and provide electricity to over 1.1 billion people who

currently lack access to electrical power [1]. To meet future energy needs across power generation, commercial and residential buildings, transportation, and industrial applications, it is essential to increase the integration of sustainable and renewable energy systems (RESs) and explore innovative alternative fuels [2] such as solar, wind, wave, and bioenergy. Photovoltaic (PV) and wind turbine (WT) installations are the most promising means of providing power in remote and rural regions [3]. However, a common drawback of these energy sources is their inherent unpredictability. Additionally, the variability of solar and wind energy often does not align with the timing of energy demand [4]. The primary challenge in integrating renewable energy into conventional energy systems is the significant requirement for large-scale energy storage devices to mitigate the variability of renewable supplies [5], [6]. These drawbacks pose significant reliability challenges in the design and operation of PV and WT systems. The recent increase in fossil fuel prices has heightened the competitiveness of green technology compared to traditional methods, thereby creating a conducive environment for their integration into microgrid systems [7].

Microgrids also play a vital role in enhancing customer reliability and resilience during network outages. The use of microgrids as fundamental building blocks in the development of smart grids represents a viable approach to decarbonizing the energy system. They are designed to withstand electrical and thermal loads [8]. In [9], The authors have proposed determining the optimal location of an Flux-coupling type superconducting fault current limiter (FSFCL) in the large-scale and multigenerator IEEE New England 39-bus test system. In [10], the authors have presented the integration of renewable energy with energy storage systems (ESSs). Additionally, they have conducted an extensive review on the optimal allocation and control of energy storage systems (ESS). In [11], the author has presented the hybrid wind-solar and wind-storage power, offering a comprehensive examination of various energy storage systems (ESSs) that are currently under development or already in operation. This includes outlining their operating principles and providing a comparative analysis of key features, technical attributes, and economic aspects. In recent years, significant progress has been made in the development of energy storage systems (ESSs) that utilize sustainable energy to meet the demands for enhanced energy efficiency and reduced emissions [12].

A variety of energy storage systems (ESSs) are available, including mechanical energy storage, thermal energy storage, electrical energy storage, and electrochemical energy storage. Options for mechanical energy storage include flywheels, compressed air systems, and pumped hydro storage [13]. Chemical energy can be converted into electrical or thermal energy through techniques known as thermochemical methods [14]. This type of storage technology relies on a dependable chemical reaction system. Most thermochemical energy systems are still in the research and development phase, facing challenges related to safety, economics,

efficiency, and other factors. Examples include fuel cells and hydrogen energy storage [15].

The incorporation of hydrogen energy storage into hybrid energy systems is progressively growing across various applications. Moreover, Power-to-hydrogen-to-power systems, based on fuel cell (FC) technology, are considered promising energy sources. They are valued for their cleanliness, lack of pollution, and efficiency. Additionally, they have the capability to store a greater amount of calorific energy in the form of hydrogen compared to most other materials [16], [17], [18], [19]. Furthermore, it operates effectively for extended durations [20]. Moreover, fuel cell (FC) techniques can be employed within microgrid systems to stabilize grids during periods of low production from renewable energy sources (RESs). Integrating these technologies with renewable energy resources enables them to function as storage devices, capable of storing energy efficiently, particularly in the presence of abundant renewable energy resources [21]. Given a sufficient supply of hydrogen, the fuel cell option offers considerably longer uninterrupted operating times compared to a battery storage system [22]. In [23], the authors have been proposed a hybrid power system consisting of PV, FC, supercapacitor (SC), and grid-connected battery energy storage system (BESS), an energy management method based on adaptive partial fuzzy sliding mode control (AFFSMC) is suggested. Muritala et al. in [24], the authors have been examined the performance effectiveness of a microgrid containing solar, wind, and a proton exchange membrane (PEM) cells with H₂ storage. Moreover, they indicated that H₂ storage is effectively used to manage the variability in load demand and RESs. A standalone hybrid microgrid with hydrogen storage can reduce fuel costs by 41% and line losses by 12% [25]. Additionally, the heat generated from the fuel cell can provide the heat needed for H₂ desorption, eliminating the need for an external heat source [26]. Many researchers considered PV and FC into studies as main and auxiliary sources. Moreover the SC and BESS were used as energy storage element to meet the load needs [27]. In [28], The authors have suggested implementing a hybrid power network (comprising PV panels, batteries, and the grid) for King Abdullah Campus at the University of Azad Jammu and Kashmir, Muzaffarabad, Pakistan. In [29], The authors have demonstrated a system aimed at providing an algorithm for EV charging and discharging that effectively tackles the challenge of peak power demand. In [30], the authors have been discussed several characteristics of modern smart grids; energy efficiency and emissions reduction, and hydrogen technology is critical for power generation, energy management, ESSs, FCs and other applications. Compared to other fossil fuels, hydrogen can establish a more environmentally friendly and sustainable energy system that can address environmental issues [31]. The economics of the energy supply networks that support smart grids can be improved by hydrogen storage systems [32]. In addition to generating energy, hydrogen can be created chemically and stored as a

gas, liquid, or carbon-based material [33]. There are benefits to converting sustainable chemical energy for smart grids, particularly as hydrogen storage technology can be very useful in off-grid applications [34]. Numerous researchers are investigating optimal hybrid systems to enhance microgrid stability under varying levels of RESs injection. In [35], Zervas et al. the authors propose that integrating PV with a hydrogen system offers greater advantages compared to standalone PV systems. This synergy enables efficient storage of solar energy by converting it into hydrogen. Hwang et al. in [36] developed a mathematical model for a PV and Hydrogen hybrid system aimed at optimizing the utilization of RESs. Their setup consisted of PV panels, an electrolyzer, a hydrogen storage tank, and FC generator. The FC generator served as a backup to supplement power demands during periods of insufficient PV generation, particularly in low solar radiation conditions. The researchers concluded that the system effectively matched the daily energy requirements of a standard residential building. To enhance the performance of the hybrid system based on a PV, FC, Battery and supercapacitor system, Samson et al. introduced an optimal load distribution strategy [37]. In their research, they employed the FC to incorporate the sporadic output from the PV system, while the battery storage was employed to address a portion of the momentary peak energy demand that the PV and FC couldn't fulfill, thereby preventing the need for excessive FC capacity. In [38], Leva and Zaninelli presented a hybrid system combining PV, wind, and diesel to provide power to a distant load. The outcomes demonstrated that the power equilibrium was maintained, and the system functioned effectively with the FC. Incorporating PV, wind, and FC technology enhanced the project's sustainability, taking into account its environmental impact. In [39], Silva et al. conducted an economic evaluation and optimization of a PV/FC/Batteries hybrid system designed for deployment in the Amazon region of Brazil. The study demonstrated that the PV-FC-battery setup, which harnessed energy stored in the form of hydrogen, was technically viable even in the challenging climatic conditions of the Amazon. In [40], Ganguly et al. developed a model and conducted an analysis of a power system integrated into a greenhouse, which included PV panels, an electrolyzer bank, and Polymer Electrolyte Membrane FC stacks. In [41], Eroglu et al. designed a portable eco-friendly dwelling that utilized a hybrid power system comprising PV panels, wind turbines, and an FC. Their system featured an 800 W PV module, a 1 kW wind turbine, and a 2 kW FC, capable of generating the electrical power of 3.8 kW. Any surplus energy produced was transformed into hydrogen through an electrolyzer and stored in a hydrogen tank.

Numerous electrification initiatives rely on the implementation of Hybrid Optimization Model for Electric Renewables (HOMER Pro) software. In [42], Karakoulidis et al. utilized HOMER software to create a model for an independent PV-DG-BT-FC system aimed at fulfilling the electricity

requirements of a machinery laboratory situated in Kavala, Greece. The simulated results demonstrated that replacing diesel generators and batteries with hydrogen was a viable technological option. This substitution not only reduced emissions, noise, and dependence on fossil fuels but also enhanced the integration of RESs.

HOMER Pro and the particle swarm optimization algorithm are commonly employed tools and techniques. In [43], the lowest Net Present Cost (NPC) and Cost of Energy (COE) can be used to select the optimal configuration. In [44] the authors have been proposed an optimal Hybrid Renewable Energy System (HRES) model incorporating PV/WT/ diesel generator (DG)/ battery (BT) for off-grid rural electrification in Bangladesh. An optimal Levelized Cost of Energy (COE) was determined for five different locations worldwide [45]. Through the application of Genetic Algorithm (GA) and HOMER, a mathematical model has been developed to ascertain the optimal size of a HRES incorporating PV/WT/BT for an isolated area [46].

The load demand of a typical microgrid system fluctuates on an hourly basis. As the load demand curve rises and falls, utilities set various rates at different times, which is known as time-of-use tariffs. The two forms of load that compose the hourly demand of the microgrid system are elastic and inelastic. Demand-side management (DSM) moves elastic loads from peak load hours to hours when the utility charges less, reshaping the whole demand model based on demand-price elasticity. Thus, customers reduce their load consumption during peak hours or shift loads to off-peak times [47]. DSM is essentially a flexible method that utilities can employ to reduce peak load demand and reduce costs [48], [49]. Utility companies typically offer incentives to customers to get them involved in the process. The six different types of load shaping strategies used by DSM include valley filling, peak clipping, load shifting, strategic conservation, strategic load increase, and variable load shape [50]. The two primary forms of DSM are incentive-based and price-based. Customers can regulate their energy consumption in the price-based DSM based on the energy price (dynamic tariff). While in the incentive-based DSM, utility cartels may group together some controlled loads based on levels of demand or generation [51]. Designing an effective DSM technique is fundamentally required to easily include the various sorts of loads. Consumers may have more freedom if the operating and emission costs of the supplied energy sources are reduced [52]. By combining DSM with various tactics including rewards, electricity prices, and penalties, the electricity consumption can be drastically reduced [53]. DSM may be able to address the microgrid issue with frequency regulation mentioned above. Demand side device turning on and off with less expensive energy usage is presented by Ozturk et al. [54].

In this study, HOMER software is used to offer an ideal planning strategy based on different resources for

a residential home. This research proposes a 100% RES and diesel generator-based optimal planning method for a residential home. According to renewable energy resource (RER) potential in the study region and electrical load demand, the optimal resource combination is selected. Since the house is off the grid, its resources should be able to meet all of its energy needs. The results for various operational reserve and unmet electric load modes are examined and numerous solutions are categorized based on cost and reliability needs. Finally, based on the system's smallest net present cost, the ideal HRES configuration is recommended. However, unlike other research, this procedure incorporates the effect of a DSM strategy, aiming to achieve optimal values for capacity shortfall. The load has been calculated accordingly. Real power consumption data for a single-family home in the Egyptian cities of Qena and Hurghada has been taken into consideration as a case study. The rest of the paper is separated into the following categories: system components are explained in Section II. The load profile, in Section III. Material and Methodology in Section IV and study location details in Section V, Homer pro software tool in Section VI, economic analysis in Section VII, Simulations results and discussion in Section VIII, System Analysis in Section IX and Conclusions in Section X.

II. SYSTEM COMPONENTS

The system's major goal is to provide hydrogen and energy at the required level of reliability with economical cost. The study integrates a microgrid that encompasses various components, as illustrated in Fig. 1. These components include DG, PV system, WT, converter, hydrogen-to-power-to hydrogen system, batteries, and loads. Besides, power-to-hydrogen-to-Power system consists of PEM electrolyzer, hydrogen tank, and fuel-cell. Hence, PEM electrolyzer employs a solid polymer electrolyte to facilitate the electrochemical reaction, and the conversion of electrical energy into hydrogen production. This type of electrolyzer is recognized for its high efficiency, quick response time, and adaptability to operate under varying load conditions, which make it well-suited for dynamic energy systems like microgrids. The system configuration is consisted mainly of three primary buses namely: hydrogen (H_2), DC power and AC power, as shown in Fig. 1. The AC and DC power buses are connected via a power converter. Here, the AC power is generated by the DG and the WTs while the DC power is obtained from PV and HES. The power converter can also convert some excess DC power to AC power, which makes up the third source of AC power. In this regard, there will be three sections of the total AC power generated which are utilized to supply the residential energy demands, and to provide the required energy for the electrolyzer in order to produce H_2 for fuel cell operation. Finally, the system's DC bus obtains the third component after it has been converted to DC power. To simulate system operation and control, the hybrid microgrid is investigated as illustrated in Figure 1 using HOMER Pro software.

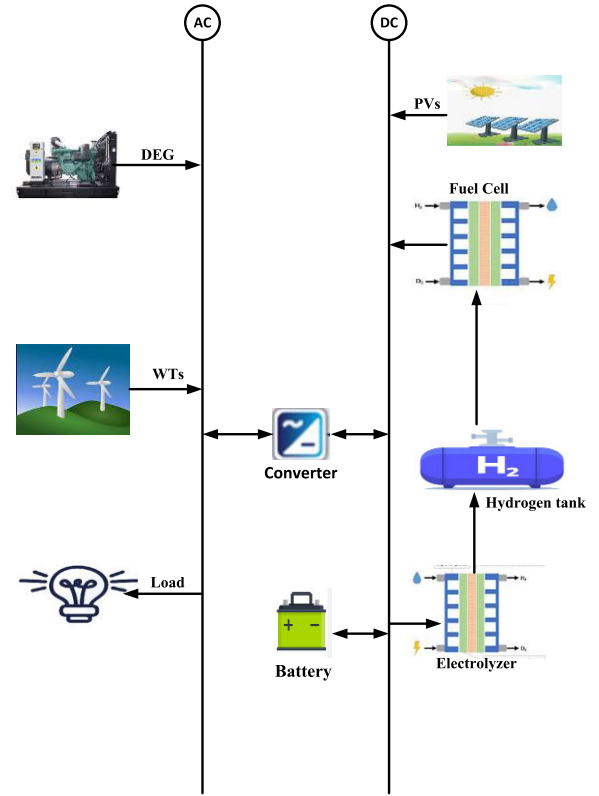


FIGURE 1. Description of hybrid microgrid system.

A. DIESEL GENERATOR MODELING

The DG acts as a backup source in cases where the RESs and storage system are insufficient to meet the energy demands in various applications, including residential, commercial, industrial, and construction sites to achieve the balance between the generation and load consumption. In [55], the authors described the amount of fuel which DG used. In [56], the amount of fuel consumed by the generator is calculated by Eq. (1) in which an auto-sizing generator was used in this study. Auto-sizing a generator entails determining the optimal generator size based on the specific power requirements of the equipment or facility it will serve.

$$F = F_0 Y_{gen} + F_1 P_{gen} \quad (1)$$

where F is the fuel consumption, F_0 is the generator's fuel curve intercept coefficient (L/h/kW), Y_{gen} is the rated capacity (kW), F_1 is the generator's fuel curve slope (L/h/kW), and P_{gen} for output (kW). Figure 2 indicates the fuel curve for the DG according to the previous equation while its efficiency curve is described in Figure 3.

B. PV PANEL MODELING

The electrical power produced from the solar PV panels can be used to determine the output power of a PV module [57].

$$P_{output} = Y_{pv} f_{pv} \left(\frac{\bar{G}_T}{G_{T,STC}} \right) [1 + \alpha_p (T_c - T_{c,STC})] \quad (2)$$

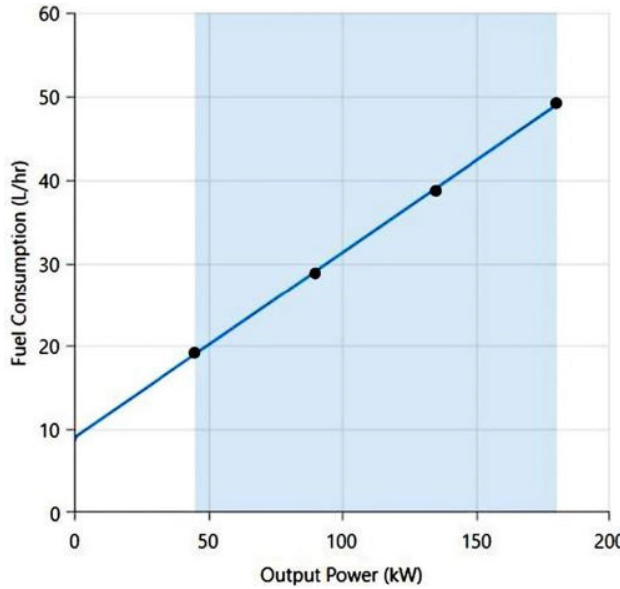


FIGURE 2. Fuel curve for DG.

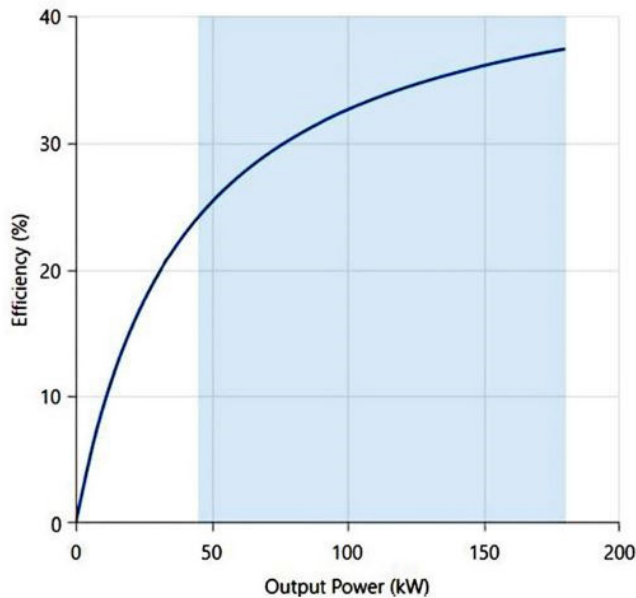


FIGURE 3. Efficiency curve for DG.

As, Y_{pv} (kW) represents the rated capacity or output power of the PV array under standard test condition, f_{pv} represents the derating factor of the PV array(%), $\bar{G}_T$ is the incident solar radiation on the PV array (kW/m^2), $G_{T,STC}$ denotes the incident solar radiation under standard test conditions, which is typically standardized as (1 kW/m^2), α_p refers to the temperature coefficient of power ($^\circ\text{C}$), T_c represents the PV cell temperature ($^\circ\text{C}$) and $T_{c,STC}$ specifies the cell temperature of the PV under standard test conditions (25°C).

C. WIND TURBINE MODELING

WT is responsible to convert the kinetic energy of the wind into mechanical energy, which is then used to generate

electricity. Hence, Eq. (3) indicates the estimated mathematical model of the WT.

$$P_{WT} = \begin{cases} 0 & v_{out} < v < v_{in} \\ \left(\frac{v - v_{in}}{v_n - v_{in}} \right) P_m & v_{in} < v < v_n \\ P_m & v_n < v < v_{out} \end{cases} \quad (3)$$

where v represents the speed of the wind (m/s), v_{in} is the cut-in speed of the wind, v_{out} is cut out speed of wind, v_n is the rated speed of the wind, and P_m is the rated power of the WT. When the wind speed is either lower than cut in speed or higher than cut out speed, the WT power (P_{WT}) is zero. If the wind speed is less than the rated speed, the WT power is directly proportional to the wind speed [58].

D. CONVERTER MODELING

In the context of RESs, a bidirectional back-to-back converter is used to convert the energy generated by renewable sources into a form that can be utilized in both islanding mode and when integrated into the utility grid. Thus, the converters play a crucial role in managing and optimizing the energy flow and ensuring compatibility between different energy sources and loads. The applied converter in this analysis has the following characteristics such as a 1 kW capacity, lifetime 15 years, inverter input efficiency is 90%, and rectifier input efficiency is 85% [59]. Here, Eq. (4) can be used for determining the inverter's output power P_{out} which is dependent on the input power P_{in} and its efficiency η_{inv} [60].

$$P_{out} = P_{in} \times \eta_{inv} \quad (4)$$

E. BATTERY MODELING

A battery is a device that stores electrical energy in the form of chemical energy and releases it as electrical energy when it is needed. Batteries are an essential component of RESs, which have the capability to acquire efficient energy management, improve system reliability, and increase the integration of RESs into the electrical grid under their generation uncertainties. The selection and sizing of batteries are dependent on the specific energy storage requirements and the characteristics of the RESs. In this analysis, the lead acid batteries are used with 1kWh capacity. Eq.(5) can be used to calculate the battery's state of charge (SOC %) [61].

$$SOC(t) = SOC(t-1) \times \int_{t-1}^t \frac{P_b(t) \times \eta_{bat}}{V_{bus} * I_{base}} * 100\% \quad (5)$$

where the battery's load power (kW) is indicated by P_b , which is calculated using Eq. (6) [62], the bus voltage is given by V_{bus} , I_{base} is the base current, and the battery's efficiency is indicated by η_{bat} (%).

$$p_b(t) = \frac{k \times Q_1(t) \times \exp(-k) + Q(t) \times k \times c \times (1 - \exp(-k \Delta t))}{1 - \exp(-k \Delta t) + c \times (k \Delta t - 1 + \exp(-k \Delta t))} \quad (6)$$

Here $Q_1(t)$ represents the total energy at the start of the operating interval and above the minimum SOC , k stands for the constant energy storage rate, c donates the storage capacity ratio, Q_t offers the total energy at the start of the operating interval, and Δt is the time interval.

F. POWER-TO-HYDROGEN-TO-POWER SYSTEM

Hydrogen is a promised energy carrier that can be produced by several methods especially splitting water molecules into hydrogen and oxygen molecules in an electrolyzer. The process is also referred to as electrolysis, and the chemical processes that occur require electricity as an energy source. After that, the produced hydrogen can be preserved until it is transformed to fulfill end-use energy demands. If the end-use require electrical energy demands, the conversion can be carried out using a FC, in which hydrogen molecules interact with oxygen to generate water and electrical energy. As a result, the power-to-hydrogen-to-power conversion by Hydrogen energy storage system (HESS) is divided into three stages:

- **Stage 1** (hydrogen production): by using the electrolyzer, electrical energy is converted to chemical energy.
- **Stage 2** (hydrogen storage): chemical energy is stored as hydrogen in hydrogen tank (Htank).
- **Stage 3** (hydrogen consumption): by using the FC, chemical energy is converted to electrical energy.

As consequently, HESS is divided into three components which are investigated in details in the following subsections.

1) HYDROGEN PRODUCTION

An electrolyzer plays a crucial role in the production of hydrogen from water through electrolysis by using electrical energy provided from RESs to split water (H_2O) into its constituent elements; hydrogen (H_2) and oxygen (O_2), through a process called electrolysis. It is commonly used for the production of hydrogen gas, which can be used as a clean and versatile energy carrier. Electrolyzers facilitate the generation of clean hydrogen for various applications such as FC vehicles, energy storage, renewable energy integration, and industrial processes, which accomplishes with the energy transition strategies towards a sustainable and low-carbon energy systems. Moreover, the hydrogen produced from electrolyzers can be used as a clean fuel for transportation, as a feedstock for chemical processes, or as a means of storing excess renewable energy. Eq.(7) represents the electrolyzer's power consumption [2], [63].

$$P_{EZ} = \frac{\dot{m}_{H_2} HHV_{H_2}}{\eta_{H_2}} \quad (7)$$

where, P_{EZ} is the electrolyzer's DC power source, η_{H_2} represents the electrolyzer's overall efficiency, $\dot{m}_{H_2}$ is the electrolyzer's mass flow rate of hydrogen (kg/s), HHV_{H_2} refers to the heating value of hydrogen fuel (MJ/kg).

2) HYDROGEN STORAGE

A hydrogen tank, also known as a hydrogen storage vessel or hydrogen cylinder, is a container designed to store hydrogen gas under high pressure. Hydrogen tanks are a crucial component in various applications that utilize hydrogen as a fuel or energy carrier. Moreover, they have different types, including compressed gas cylinders and cryogenic tanks. Compressed gas cylinders are made of high-strength materials, such as carbon fiber-reinforced composites or steel, and store hydrogen at high pressures (usually between 350 and 700 bar). Cryogenic tanks, on the other hand, store hydrogen in a liquid state at extremely low temperatures (-253°C) and are used for large-scale storage. The pressure of hydrogen (Q_{ht}) in the hydrogen tank can be calculated by using Eq.(8) [64].

$$Q_{ht} = \left(\frac{R \times T}{V_{ht}} \right) \times n_{ht} \quad (8)$$

where n_{ht} refers to the number of moles in the tank, V_{ht} refers to the volume of the hydrogen tank, R is the universal gas constant, and T is the temperature.

3) HYDROGEN CONSUMPTION

Hence, the FC converts the chemical energy of a fuel, typically hydrogen, into electrical energy through a process called electrochemical reaction. FCs exhibit various prominent features such as high efficiency, low emissions, and versatility, which make them a promising technology for various applications. Besides being also environmentally friendly as they produce electricity without combustion, resulting in minimal emissions of greenhouse gases and pollutants. As research and development continue, FCs are expected to play a significant role in the transition towards a sustainable and low-carbon energy future. Eq. (9) shows a mathematical expression for the power produced by the FC [65], [66], [67].

$$P_{FC} = N * V_{FC} * I_{FC} \quad (9)$$

N is the stack's total number of cells, V_{FC} is the FC's output voltage, P_{FC} is the FC's power output and I_{FC} is the FC's current. In addition, output voltage (V_{FC}) is defined by Eq. (10):

$$V_{FC} = E - V_{act} - V_{con} - V_Q \quad (10)$$

where, E is the internal voltage of the FC, V_Q is the ohmic voltage, V_{con} is the concentration voltage, and V_{act} is the activation voltage. The hydrogen production, consumption and storage should be included in the power balance equation as depicted in [4].

III. LOAD PROFILE

By analyzing the load demand, it is applicable to identify peak usage periods, estimate the total energy consumption, and determine the capacity requirements for energy generation and storage systems. The load demand for this single residential home was determined using power data from mature energy consumer's equipment such a refrigerator, lighting, and previous year's electricity bills as shown in TABLE 1 [68]. This study focuses on supplying power to a

TABLE 1. Average energy demand for a single home (source: RET Screen Software) [69].

Appliances	Load (W)	Quantity	Average Hourly usage/day	Energy demand		
				kWh/d	kWh/month	kWh/yr
Electric iron	1000	1	0.5	0.5	15	182.5
Fan	70	3	5	1.05	31.5	383.2
Personal Computer	200	1	2	0.4	12	146
Energy-efficient light bulbs (CFL)	40	4	10	1.6	48	584
Energy-efficient light bulbs (CFL)	15	2	10	0.3	9	109.5
Refrigerator	200	1	12	2.4	72	876
TV	150	1	5	0.75	22.5	273.7
Washing Machine	700	1	0.5	0.35	10.5	127.7
Motor Pump	370	1	1	0.37	11.1	135.05
Total	2745	18	46	7.72	231.6	2817.8

TABLE 2. Daily residential load on priority lines [69].

Appliances	HPLL kWh/day	LPLL kWh/day
Electric Iron	0.5	-
Fan	1.05	-
Personal Computer	-	0.4
Energy Saver (CFL) (40 w)	1.6	-
Energy Saver (CFL) (15 w)	-	0.3
Refrigerator	2.4	-
TV	-	0.75
Washing Machine	-	0.35
Motor Pump	0.37	-
Total	5.92	1.8

two-story residential house with numerous electrical appliances, having an average load of 370 kWh/d and a peak load of 28.11 kW. Upon implementing DSM strategies outlined in TABLE 2, the average load decreased to 284.16 kWh/d, with a peak load of 21.71 kW.

IV. MATERIALS AND METHODOLOGY

In this section, the applied technique and materials used for the analysis are covered which involves the mathematical equations, the technical and financial parameters taken as factors for the simulation, and the study areas' weather conditions. In this study, the DSM strategy is applied in areas where a hybrid system comprising solar energy, WTs, and hydrogen is utilized to generate electricity. The study evaluates the impact of this strategy on the system's NPC and its harmful emissions, which affect the environment. Due to its rapid and smart capacity to load-shrink during periods in which demand exceeds supply, DSM, or Demand Side Management, is widely utilized in smart grid systems. In situations where the energy supply falls short of the energy requirements for residential consumption, researchers are

exploring ways to administer DSM strategies to domestic appliances. The aim is to enhance energy efficiency and achieve conservation by optimizing how household devices use electricity [69]. The suggested method integrates a standalone hybrid power system design with DSM. It is expected that a smart distribution board may switch loads automatically depending on priority. Two priority load lines in the smart distribution board are low priority load lines (LPLL) and high priority load lines (HPLL). The algorithm process can be described on the following steps:

- 1- If $P_{\text{generation}} \geq P_{\text{demand}}$: all loads are always on.
- 2- If $P_{\text{HPLL}} \leq P_{\text{generation}} < P_{\text{consumed}}$: disconnect LPLL but keep connecting HPLL until $P_{\text{generation}} \geq P_{\text{consumed}}$.

Load shedding continues until LPLL's power consumption exceeds its power generation, at which point all operated lines are disconnected. The system design load has been determined to be 284.16 kWh on HPLL. According to the user demands and daily usage, the HPLL also contains auxiliary appliances such as an electric iron, fan, energy-saving appliance, refrigerator, and motor pump. Auxiliary devices

TABLE 3. Shows location in egypt.

City	Latitude (°N)	Longitude (°E)
Qena	27°10.2	32°46.7
Hurghada	27°15.6	33°49.0

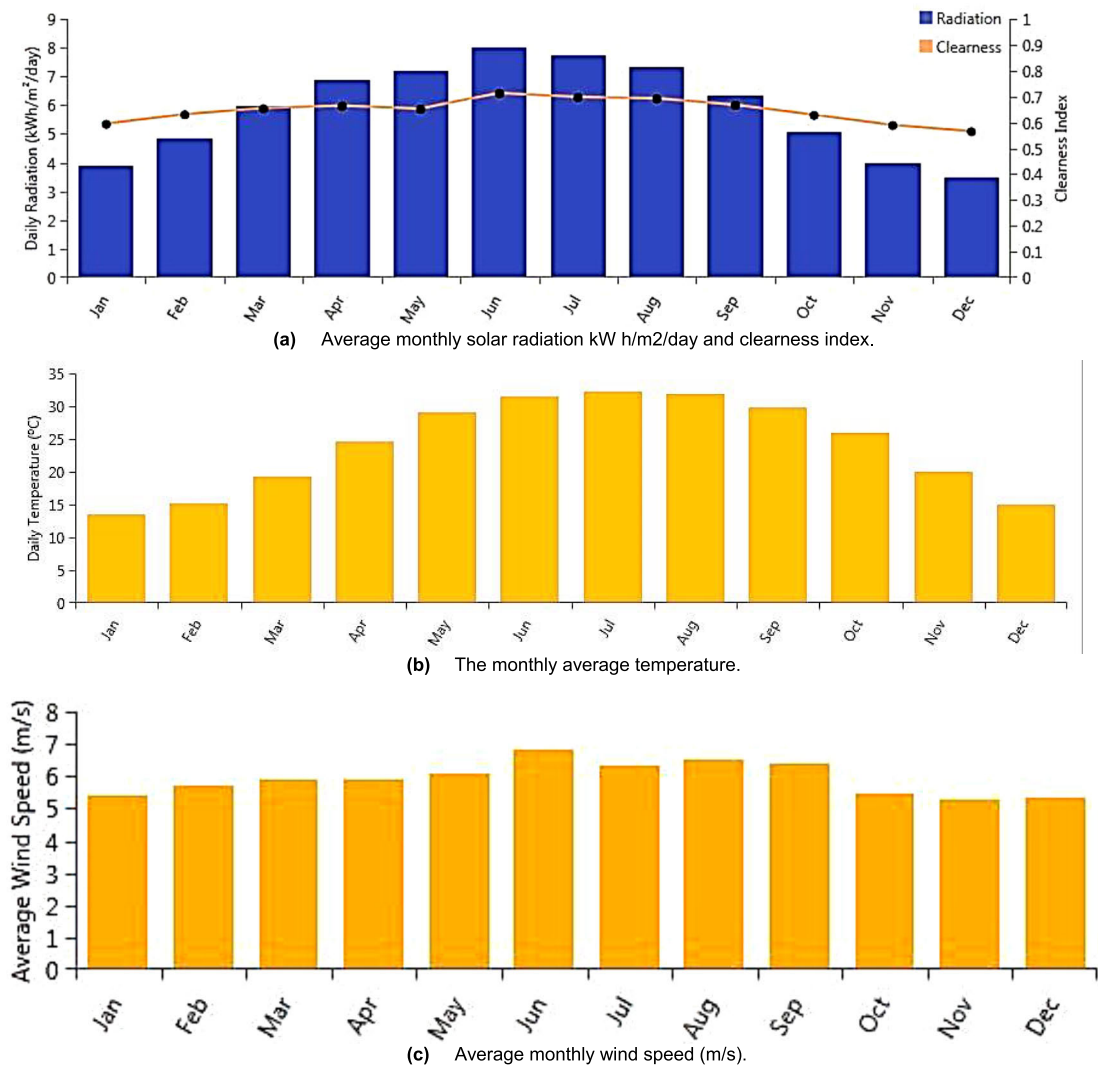


FIGURE 4. Resources assessment in qena city.

like personal computers, low-wattage energy-saving devices, televisions, and washing machines are among the LPLLs. In TABLE 2, the average energy requirements of a single-family home.

V. STUDIED LOCATION

The current study was carried out in two cities in Egypt. All with different weather conditions. TABLE 3 indicates the study’s areas’ geographic coordinate system. The global horizontal irradiance (GHI), monthly average wind speed, and average monthly temperature were only a few of the variables included in the weather data. Such data was taken from the weather data center module at NASA.

In Qena city, Figure 4 (a) indicates the solar radiation during the period of a year the annual average GHI is measured at 5.89 kWh/h/day. As shown in Figure 4(b), the average temperature is 13.49°C in January and 32.72°C in July, with January having the lowest average temperature and the annual average temperature is 24°C. Hence, evaluating the climatic factors that affect energy production and system performance, depends on knowing how the temperature varies throughout the year. The annual average wind speed is 5.93 m/s. While Figure 4(c) reveals that the highest average wind speed is 6.82 m/s in June.

In Hurghada city, Figure 5(a) shows the amount of solar radiation during a year. The average GHI for the year is

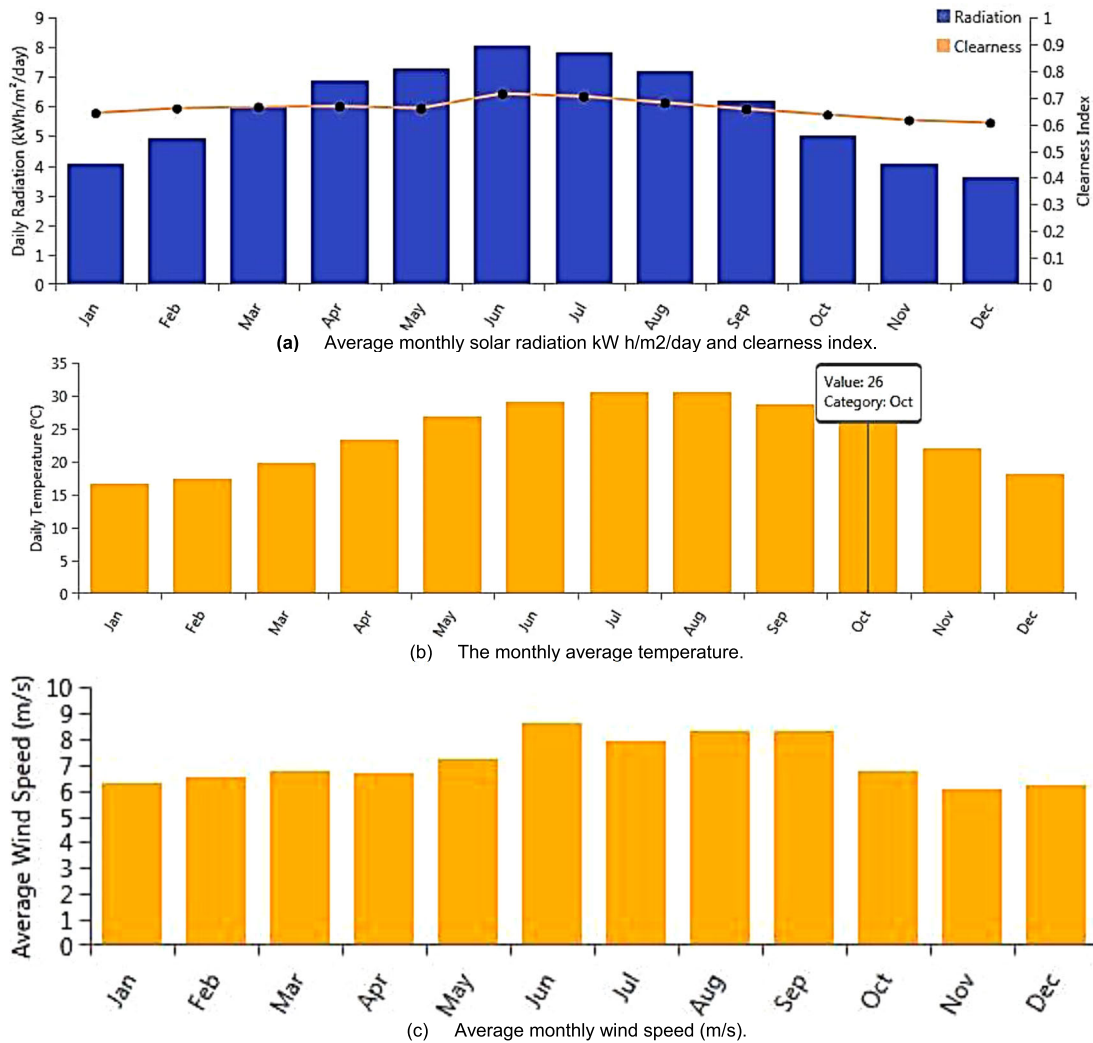


FIGURE 5. Resources assessment in hurghada city.

5.93 kWh/kw/day and the annual average temperature is 24.11°C. The highest temperature in is 16.69°C in January and lowest temperature is 30.6°C in August as depicted in Figure 5(b). The lowest value of wind speed is 6.23 m/s in December and the highest value of wind speed is 8.63 m/s in June, as shown in Figure 5(c). In addition the annual average wind speed is 7.13 m/s.

VI. HOMER PRO SOFTWARE TOOL

HOMER Pro is a software tool used for the analysis, optimization, and simulation of hybrid RESs. It allows users to design and optimize hybrid power systems by considering various components such as solar PV, WTs, batteries, generators, and grid connection. It helps determine the optimal system configuration, sizing, and dispatch strategy based on specific energy requirements and constraints. Additionally, it assesses the project's sensitivity analysis and economic viability. The software provides tools for analyzing RESs, such as solar irradiation and wind speed data. It helps to

assess the availability and variability of these resources at a specific location, allowing for accurate system modeling and performance predictions. The flowchart for the HOMER software is shown in Figure 6. [70]. The HOMER software has two different control strategies:

load following (LF) and cycle charging (CC). As a result, generators operate under the LF approach only when essential to fulfill load requirements, with renewable energies being prioritized for tasks such as battery charging. Conversely, generators are utilized in the CC approach to simultaneously charge batteries and satisfy demand.

VII. ECONOMIC ANALYSIS

HOMER Pro calculates several financial metrics to evaluate the project's economic viability. These metrics include the levelized cost of electricity (LCOE), NPC and Levelized cost of hydrogen production (LCOH). These metrics help on evaluating the project's profitability, returning on investment,

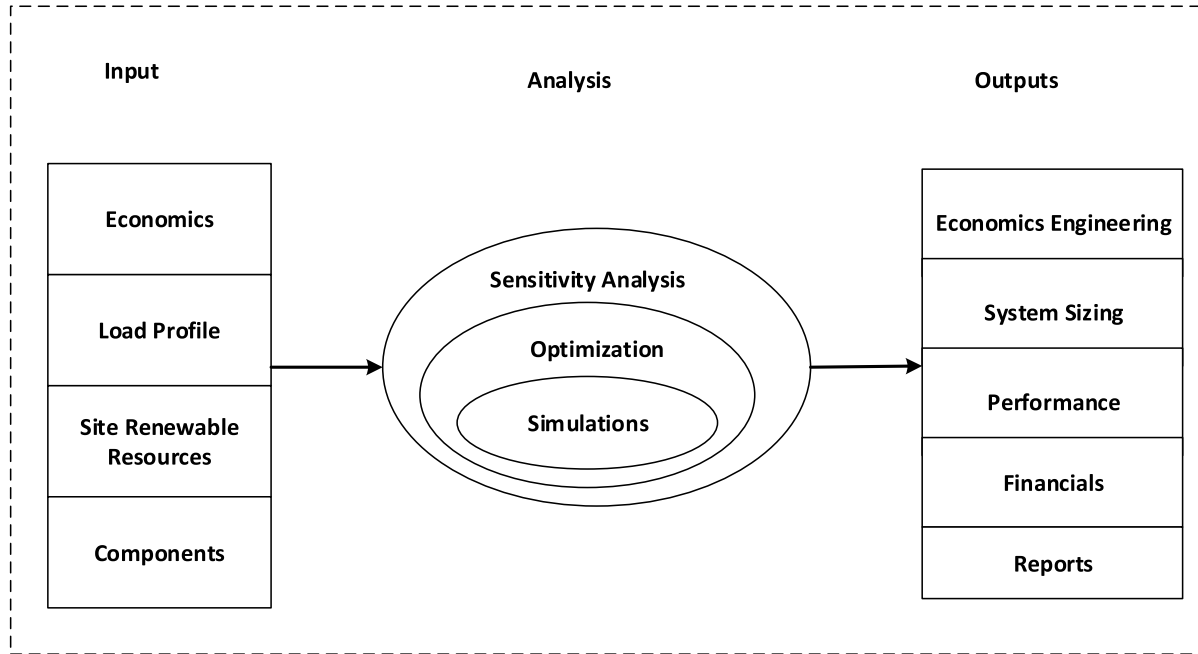


FIGURE 6. The HOMER software's flowchart.

TABLE 4. Technical and economic inputs into homer pro software.

Components	Capital Cost (\$/kW)	Replacement (\$/kW)	O&M(\$/hour)	Ref
Diesel(\$/kW)	600	500	0.025	[71]
PV(\$/kW)	1420.70	1420.70	30.20	[72]
Wind turbine(\$/kW)	1097	1097	10.97	[72]
Converter(\$/kW)	600	600	30	[59]
Battery(\$/kW)	300	300	10	[71]
Electrolyzer(\$/kW)	100	100	8	[73]
Hydrogen Tank(\$/kg)	1.5	0.5	0.6	[74]
Fuel Cell(\$/kW)	450	400	0.15	[74]

and financial sustainability. The software's technical and economic inputs are presented in TABLE 4.

A. NET PRESENT COST

The NPC is a financial metric used to evaluate the total cost of a project over its lifetime, taking into account the time value of money. It represents the sum of all cash outflows (capital costs, operating costs, and maintenance costs) discounted to their present value. The NPC in this study is determined using Eq. (11) [75].

$$C_{NPC} = \frac{C_{annual, total}}{CRF(i, R_{project})} \quad (11)$$

As, $C_{annual, total}$ is the total annual cost of the project (\$/yr); CRF is the capital recovery factor; i is the real interest rate (%); and $R_{project}$ is the project lifetime.

The CRF is a ratio used to calculate the present value of an annuity and may be calculated analytically

using Eq. (12) [76].

$$CRF_{(i,N)} = \frac{i(1+i)^N}{((1+i)^N - 1)} \quad (12)$$

As, N the number of years, i represents the real interest rate (%).

B. LEVELIZED COST OF ENERGY

The $LCOE$ is defined as the ratio of the system's average cost per kWh of valuable electrical energy produced. The $LCOE$ is shown in the following Eq. (13) [75].

$$LCOE = \frac{NPC(\$)}{\sum_{h=1}^{h=8760} P_{load}(h)(kWh)} \times CRF \quad (13)$$

VIII. SIMULATIONS RESULTS AND DISCUSSION

This paper conducts simulations on a standalone microgrid incorporating PV/WT/DG/BESS/HESS technologies in a remote village in Upper Egypt, using HOMER software to assess techno-economic advantages. The study also identifies the optimal power system configuration based on the

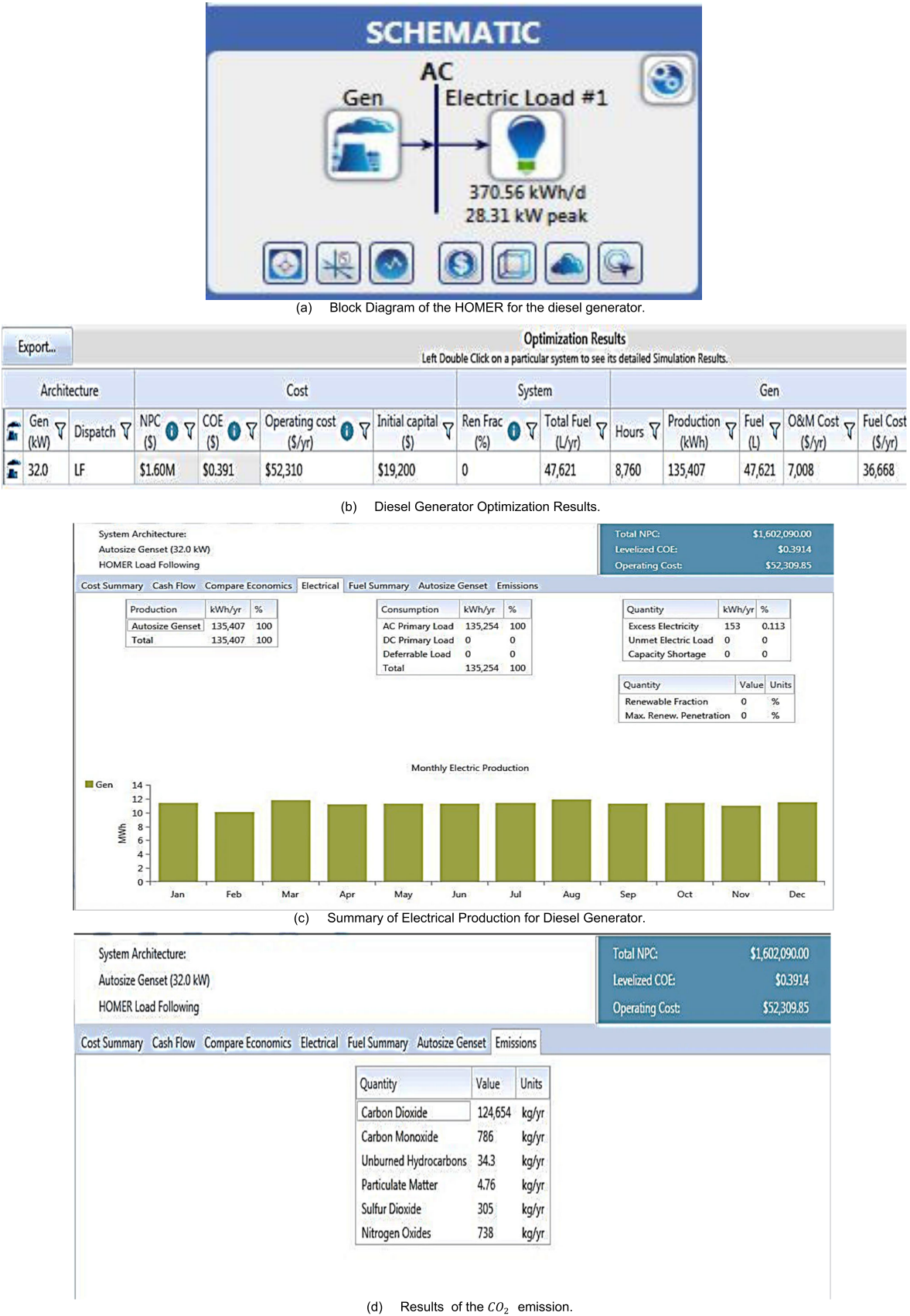
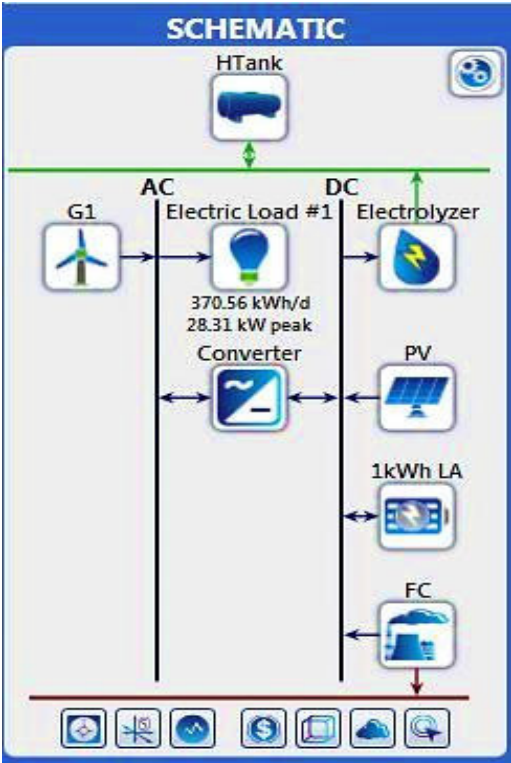


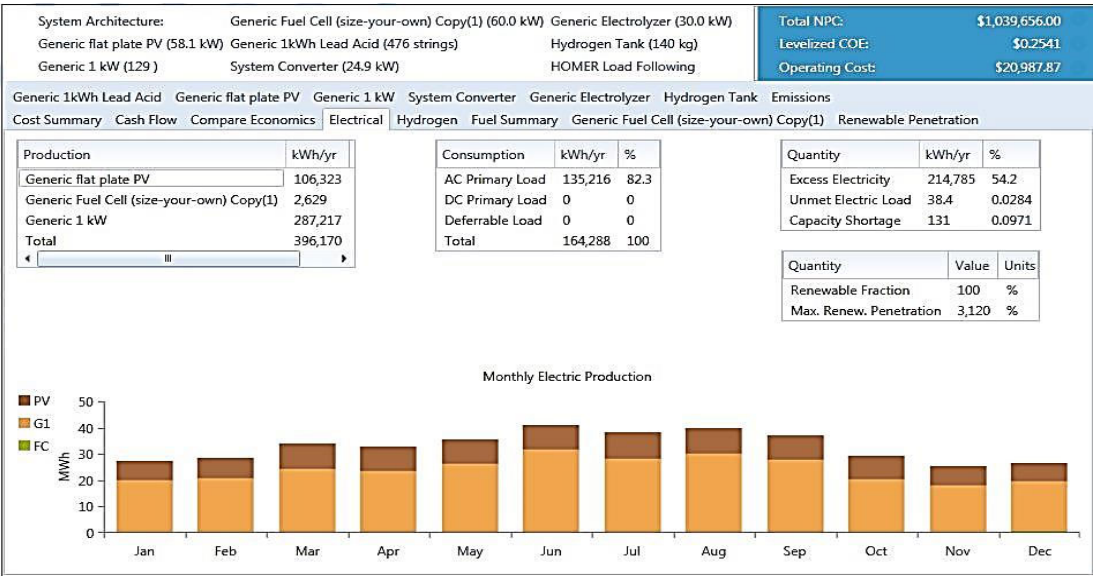
FIGURE 7. Simulation result of base case.



(a) Diagram of the HOMER for HRES without DSM.

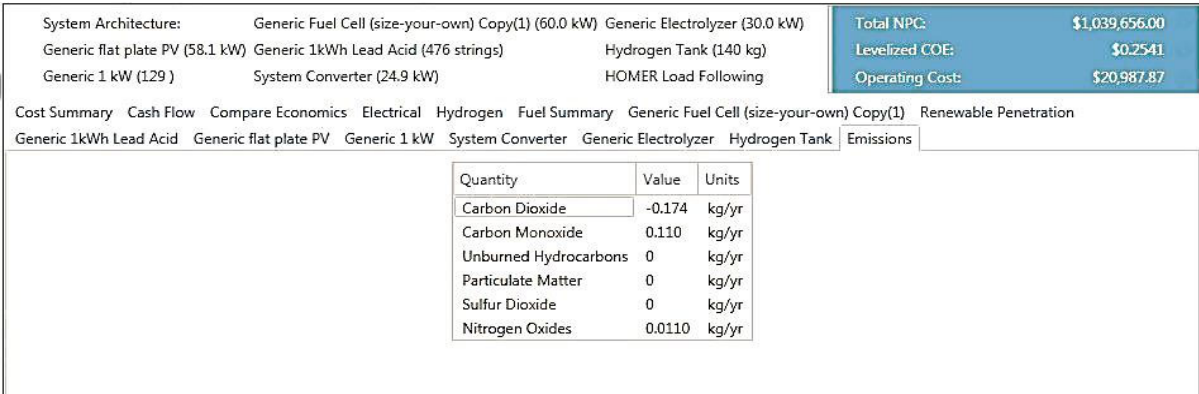
Optimization Results																	
Left Double Click on a particular system to see its detailed Simulation Results.																	
Architecture										Cost				System			
	PV (kW)	G1	FC (kW)	1kWh LA	Electrolyzer (kW)	HTank (kg)	Converter (kW)	Dispatch		NPC (\$)	COE (\$/kWh)	Operating cost (\$/yr)	Initial capital (\$)	Ren. Frac. (%)	Total Fuel (L/yr)	Hours	Production (kWh)
	58.1	129	60.0	476	30.0	140	24.9	LF		\$1.04M	\$0.254	\$20,988	\$404,566	100	552	163	2,629
	137		80.0	1,003	20.0	160	29.1	LF		\$1.60M	\$0.390	\$34,846	\$542,328	100	227	79.0	1,083
		410	60.0	637	50.0	160	62.6	LF		\$1.61M	\$0.394	\$30,429	\$691,903	100	936	282	4,456
	180	1,017	40.0		50.0	160	74.8	CC		\$2.76M	\$0.676	\$44,488	\$142M	100	4,261	2,466	20,292

(b) HRES without DSM Optimization Results.

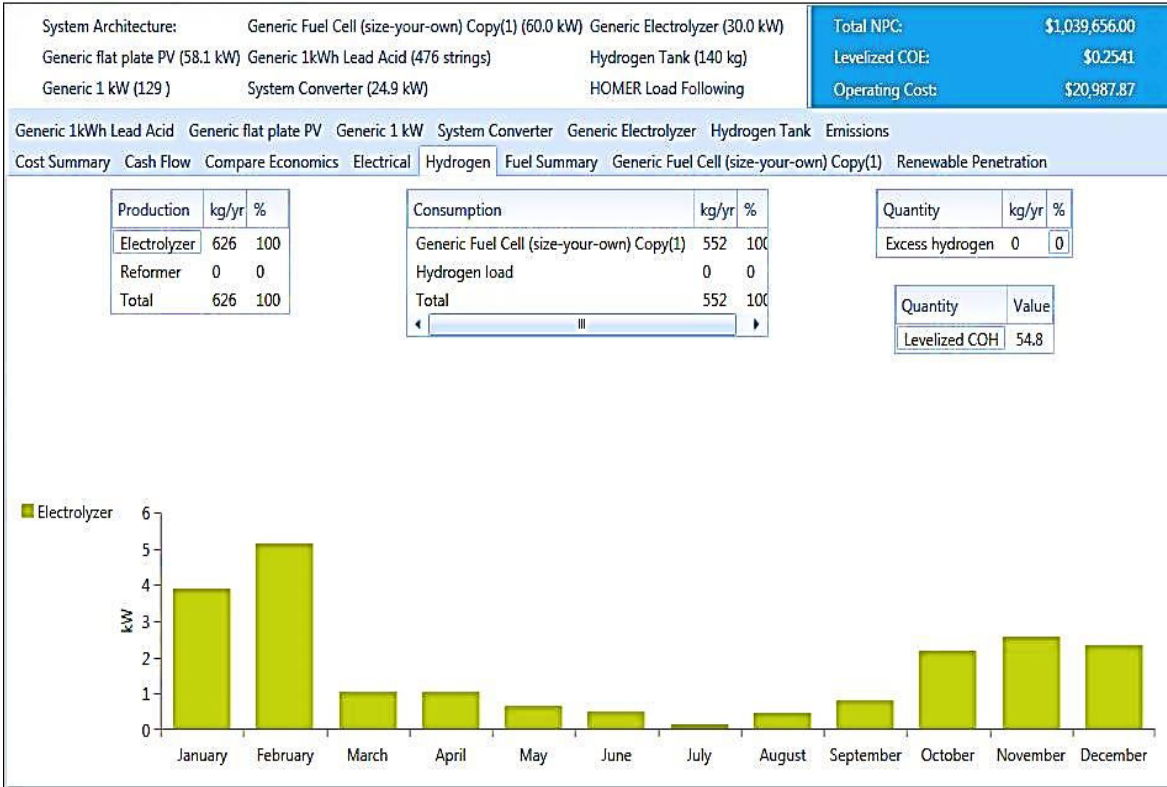


(c) Summary of Electrical Production for HRES without DSM.

FIGURE 8. Simulation result of the hybrid system in qena city without DSM.



(d) Diagram of the emission without DSM.



(e) Summary of Hydrogen Production for HRES without DSM.

FIGURE 8. (Continued.) Simulation result of the hybrid system in qena city without DSM.

cost of energy (COE) and annual net present cost (NPC). The hybrid power system is tailored to accommodate varying seasonal residential loads at the project site, aiming to provide the minimum required operational reserve for grid reliability. A comparative analysis of techno-economic and environmental benefits is presented, examining different configurations of DG, PV/WT/DG/BESS/HESS with and without DSM.

A. BASE CASE (DG ONLY)

In this particular scenario, the power requirement is solely fulfilled by a DG. The total designed electrical load is

370.56 kWh per day and 28.31 kW peak as shown in Figure 7(a), and it is assumed that the DG will supply this load. Such configurations are commonly employed in remote or off-grid locations where a reliable grid connection is either impractical or unreliable. Thus, DGs act as the primary power source in this setup. A 32 kW of DG fulfills this load with \$1.60M NPC, \$0.391 cost of energy (COE) as shown in Figure 7(b). The excess electricity is 153 kWh/yr which represents 0.113% of the total energy production resulting from the DG as the auto size genest produced 135.407 kWh/yr, as Figure 7(c) represents the electrical production summary for diesel generator. Moreover,

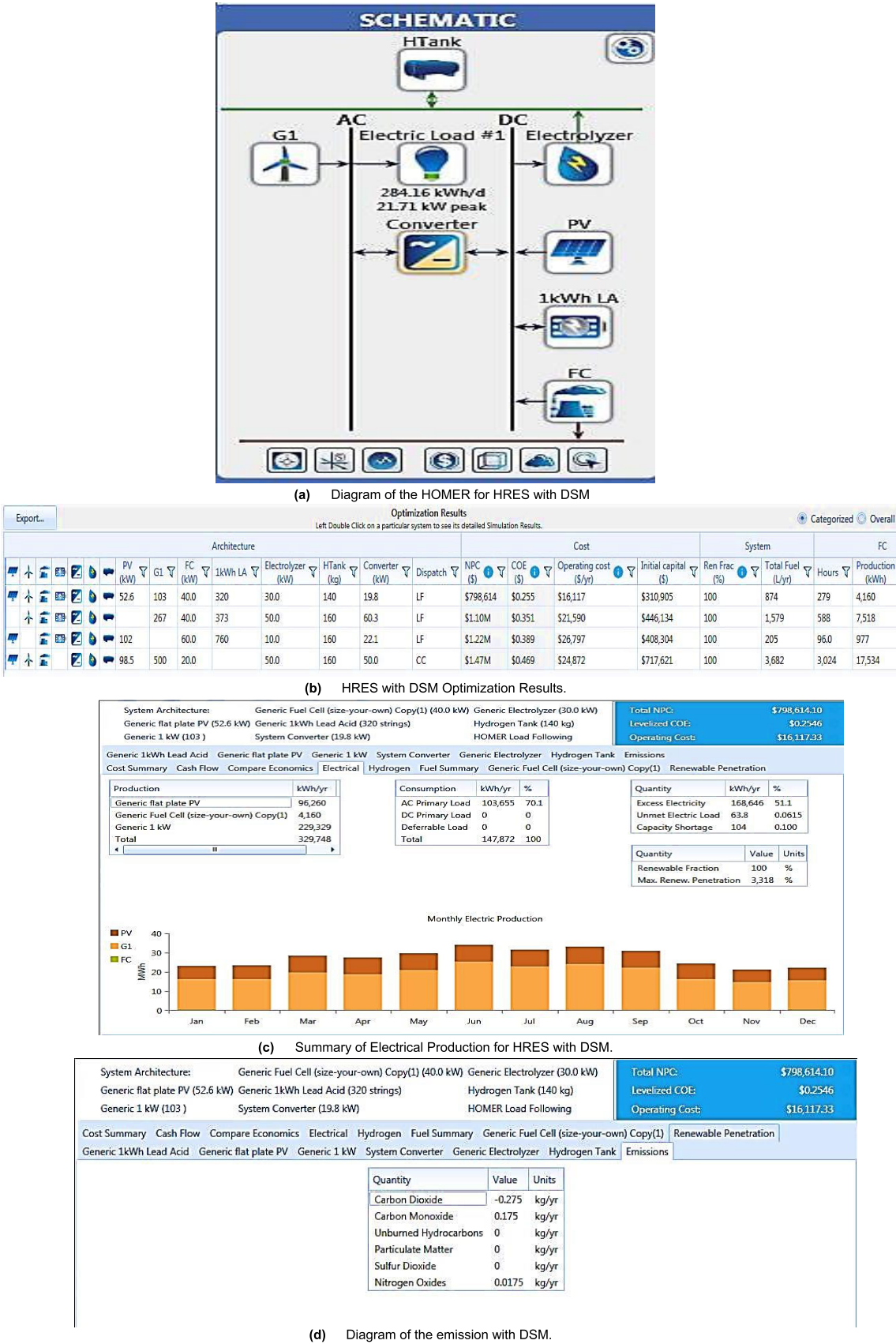


FIGURE 9. Simulation result of the hybrid system in qena city with DSM.

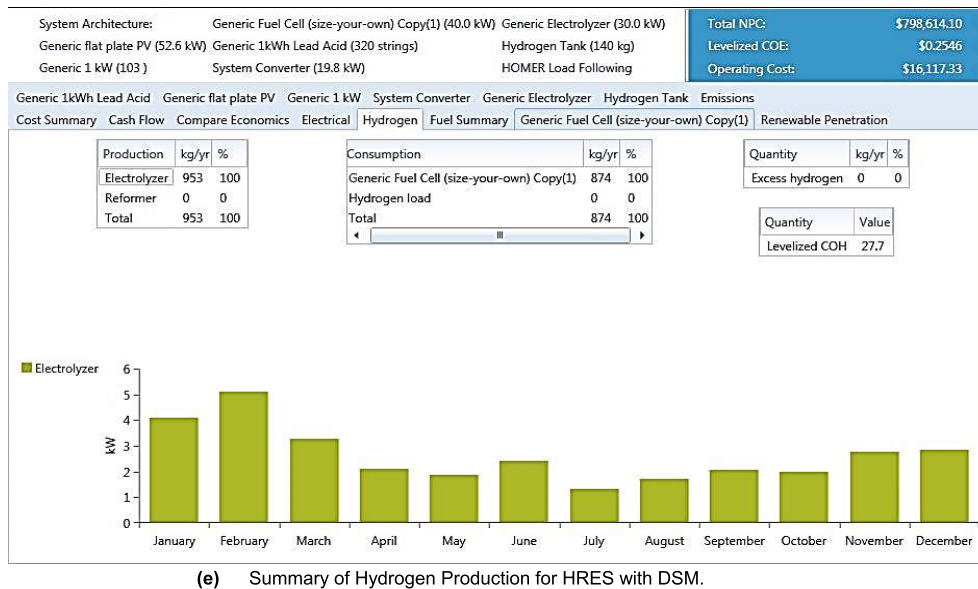


FIGURE 9. (Continued.) Simulation result of the hybrid system in qena city with DSM.

the AC load absorbed 135,254kWh/yr. Homer Pro software also indicates the value of harmful emission that produced from using DG. The carbon dioxide (CO_2) which produced is 124,654 kg/yr as shown in Figure 7(d).

B. HYBRID SYSTEM IN QENA CITY

In this scenario, the case study revolves around designing a system incorporating both a hybrid energy system and a storage system utilizing hydrogen and batteries. As mentioned earlier, the total estimated electrical load is 370.56 kWh/d. To fulfill this demand, a combination of various sources is employed, comprising a hybrid system integrating PV modules, WTs, batteries, electrolyzer, HTank, and FC, with and without DSM.

1) HYBRID SYSTEM WITHOUT DSM

The system configuration is illustrated in Figure 8(a), depicting the interconnected components working collaboratively to fulfill electricity demand. The optimal configuration of the HRES includes 58.1 kW PV, 129 kW WTs, 60 kW FC, 476 kWh lead acid (LA) batteries, 30 kW electrolyzer, 140 kg HTank, 24.9 kW converter with \$1.04M NPC and \$0.254 COE. The system strategy is LF strategy as shown in Figure 8(b). The excess electricity is 214.785 kWh/yr with 54.2% of the total power which has been produced from these configuration of microgrid as PV contributed 106,323 kWh/yr of the total electricity represents 26.84% of the total electricity power produced whereas the WTs generate 287,217 kWh/yr which represents 72.5%. Beside the FC produce about 2,629kWh/yr which represents 0.66% of the total electricity power produced, as shown in Figure 8(c). The carbon dioxide emission is nearly zero as displayed in

Figure 8(d). Further, the electrolyzer produces 626 kg/yr of hydrogen with 54.5 LCOH, as shown in Figure 8(e).

2) HYBRID SYSTEM WITH DSM

In this case study, the total planned electrical load of 370.56kWh/d has been reduced by the HPLL's load of 284.16 kWh/d and it fed by HRES as shown in Figure 9(a). HRES consists of 56.6 kW PV, 103 kW WT, 40 kW FC, 320 kWh LA batteries, 30 kW electrolyzer, 140 kg HTank and 19.8 kW converter with LF strategy. The NPC is \$798,614 and \$0.255 COE as shown in Figure 9(b). The excess electricity is 168,646 kWh/yr. with 51.1% of the total production power as PV contributed 96,260kWh/yr. of the total electricity which denotes 29.2% of the total electricity power produced whereas the WT generates 229,329 kWh/yr. that signifies 69.5% and FC provides 4160 kWh/yr. which is 1.26% of the total electricity power produced, as shown in Figure 9(c). Moreover, carbon dioxide emission is nearly zero as shown in Figure 9(d). The electrolyzer produces 953 kg/yr. of hydrogen with 27.7 LCOH, as shown in Figure 9(e).

C. HYBRID SYSTEM IN HURGHADA CITY

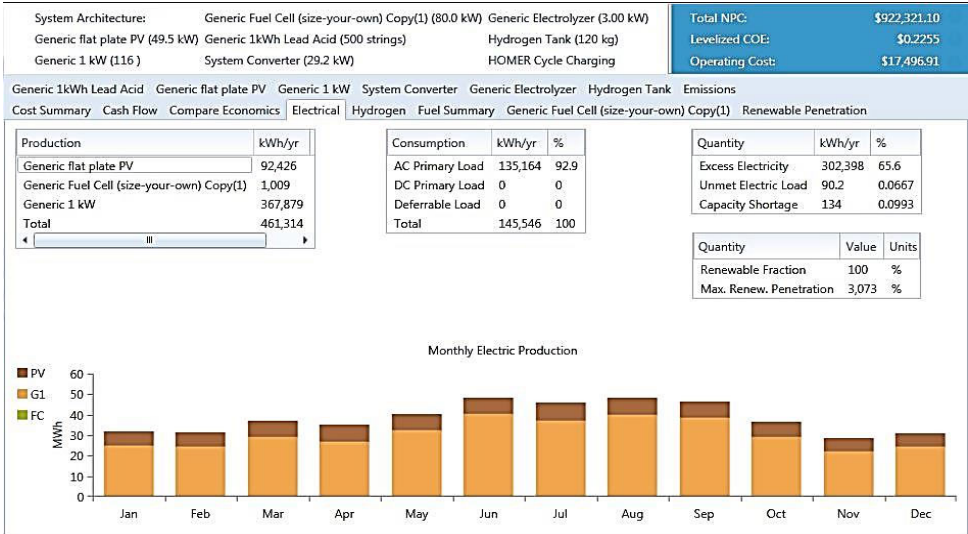
In this case study, a system is designed that includes a hybrid system and a storage system that uses hydrogen and batteries in Hurghada city. At this location, the estimated load is the same as the load and the same combination of HRES that fulfill the load as shown in Figure 8(a). In this section two different case study, Hybrid system without DSM and Hybrid system with DSM.

1) HYBRID SYSTEM WITHOUT DSM

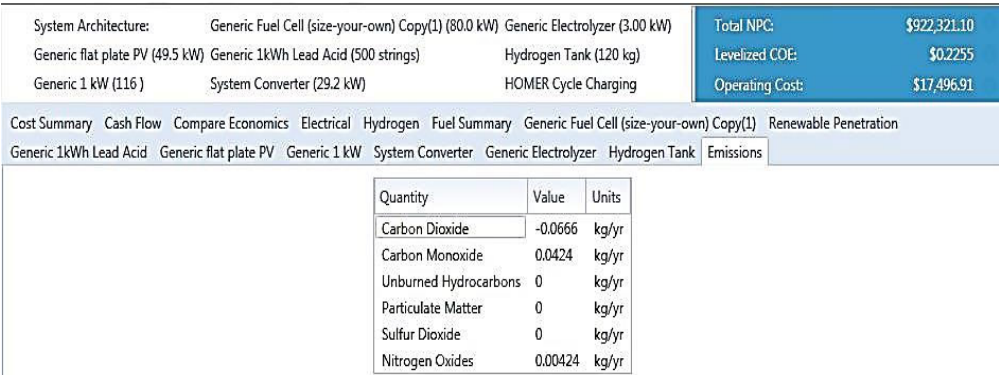
In the initial case study without employing DSM, the optimal configuration of the HRES comprises 49.5 kW of PV panels,

Optimization Results																
Left Double Click on a particular system to see its detailed Simulation Results.																
Architecture										Cost				System		
PV (kW)	GL (kW)	FC (kW)	1kWh LA	Electrolyzer (kW)	HTank (kg)	Converter (kW)	Dispatch	NPC (\$)	COE (\$)	Operating cost (\$/yr)	Initial capital (\$)	Ren. Frac (%)	Total Fuel (L/yr)	Hours	FC	
49.5	116	80.0	500	3.00	120	29.2	CC	\$922,321	\$0.226	\$17,497	\$392,867	100	212	14.0	1,009	
283	80.0	412	6.00	80.0	42.6	LF	LF	\$1.06M	\$0.259	\$19,016	\$483,549	100	348	134	1,658	
127	120	1,013	4.00	160	29.2	LF	LF	\$1.57M	\$0.384	\$33,842	\$547,522	100	157	54.0	746	

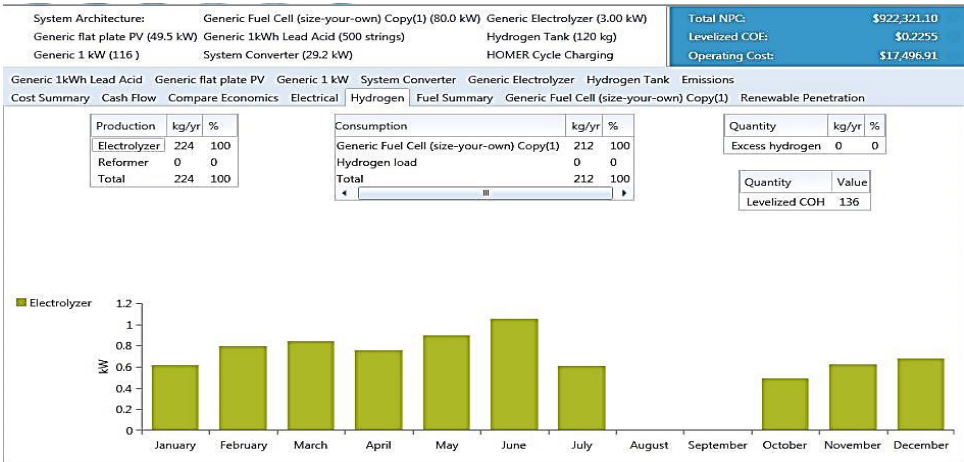
(a) HRES without DSM Optimization Results.



(b) Summary of Electrical Production for HRES without DSM.



(c) Diagram of the emission without DSM.



(d) Summary of Hydrogen Production for HRES without DSM.

FIGURE 10. Simulation result of the hybrid system in hurghada city without and with DSM.

116 kW of WT, 80 kW of FC, 500 kWh of LA batteries, a 3 kW electrolyzer, 120 kW hydrogen tank, and a 29.2 kW converter, resulting in an NPC of \$0.922 million and a COE of \$0.226 per kWh. The system strategy is CC strategy as shown in Figure 10(a). The excess electricity is 302,398 kWh/yr with 65.6% as PV contributes with 96,260 kWh/yr. of the total electricity that represents 20.9% of the total electricity power produced and WT generates 367,879 kWh/yr. which signifies 79.7% and FC produces 1009 kWh/yr that characterizes by 0.22% of the total electricity power produced, as shown in Figure 10(b). The carbon dioxide emission is recorded nearly zero value as shown in Figure 10(c). The electrolyzer produces 224 kg/yr. of hydrogen with 136 LCOH, as shown in Figure 10(d).

2) HYBRID SYSTEM WITH DSM

In second case study, the optimal configuration depends on DSM is considered, the HPLL's load of 284.16 kWh/d and it fed by the same HRES as this combination includes 30.2 kWh/yr of PV, 91 kW WT, 40 kW FC, 288 kWh LA of batteries, 9 kW of electrolyzer, 269 kg of HTank and 19.7 kW of converter with \$646,046 NPC and \$0.206 COE. The applied strategy is LF, as shown in Figure 10(e). PV produces 56,375 kWh/yr with 16.18% of the total produced power, WT generates 288,595 kWh/yr. with 28.82% and FC provides 3490 kWh/yr. with 1% of the total power generated. The value of excess power is 192,826 kWh/yr., as shown in Figure 10(f). The amount of hydrogen that produced is 850 kg/yr by electrolyzer with 25.1 LCOH as shown in Figure 10(g). The carbon dioxide emission is nearly zero as illustrates in Figure 10(h).

IX. SYSTEM ANALYSIS

The simulation results from the HOMER software are presented in this part, along with evaluations of the best configurations for each area. Operational characteristics including excess power, annual energy output, operating cost, hydrogen production, CO_2 emissions, etc. are evaluated for every simulated system at the various sites around the nation. The technological, economic, and environmental issues each represent part of the paper.

A. ELECTRICITY PRODUCTION AND ECONOMIC ANALYSIS

Electricity production and economic analysis based on a combined system of WTs, PV panels, and a hydrogen system, can be effectively assessed using the HOMER software. This integrated approach leverages RESs to generate electricity and includes hydrogen production for energy storage. The appropriate configuration for each site when feeding different load with different sources. The research results indicate that due to the weather conditions at each site differ, the configuration of the optimal system depends on the location. In the first scenario, power generation from the 32 kW generator yields 135,407 kWh/yr, as detailed in TABLES 5 & 6. Within this total production, there is an excess of 153 kWh/yr, representing 0.113% of the overall output. Economic metrics,

including NPC, COE, and operating costs, stand at \$1.60 million, \$0.391, and \$52,310, respectively, indicating relatively high costs. To minimize expenses, various Renewable Energy Sources (RESs) such as WT and PV are strategically placed in different locations. Additionally, a hydrogen storage system is implemented.

In Qena city, the most effective setup to fulfill a daily demand of 370.56 kWh, as outlined in TABLE 5, consists of a 58.1 kW PV system, 129 kW WTs, and a 60 kW FC. This arrangement yields a \$1.04 million NPC, \$0.255 COE, and an operating cost of \$20,988. Notably, the NPC and COE for this setup are lower than the base case, indicating a 35% reduction in NPC, a 0.42% decrease in COE, and a significant 59.88% reduction in operating costs. This configuration produces an approximate total electricity output of 396,169 kWh/yr. As illustrated in TABLE 6, the AC primary load consumes 135,216 kWh/yr., which constitutes 34.1% of the overall energy production. Simultaneously, the electrolyzer absorbs 29,072 kWh/yr to produce hydrogen, which is stored in the Htank and subsequently utilized by the FC to generate electricity. The electrolyzer energy consumption value accounts for 7.34% of the estimated total electricity. Furthermore, the surplus electricity has increased from its base case value, reaching 214,785 kWh/yr., which represents 54.2% of the total energy production in the current scenario. By implementing DSM, the originally planned electrical load of 370.56 kWh/d is adjusted to accommodate the 284.16 kWh/d demand of the High-Priority Load Profile (HPLL). This leads to a reduction in the electricity production from components involved in supplying loads with the High-Priority Load Profile (HLPP), such as PV, WT, and FC. Consequently, the total electricity production decreases from 396,169 kWh/yr to 329,749 kWh/yr, which achieves a reduction of 16.8%. The surplus electricity also decreases by 21.5%, now representing 51.1% of the total electricity production, as indicated in TABLE 6. Furthermore, the energy consumed by the AC load decreases and now represents 31.4% of the total energy production. Conversely, the energy consumed by the electrolyzer increases by 13.4% of the total energy production value. This reduction in load results in cost savings due to the downsizing of components. Specifically, the NPC decreases from \$1.04 million to \$0.798614 million, a reduction of 23.2%, and the operating cost decreases from \$20,988 to \$16,117, also reflecting a reduction of 23.2%. The dispatch strategy for both optimal configurations is Load Following (LF). In Hurghada city, the initial total energy production is 461,314 kWh/yr. Upon implementing DSM, there is a reduction in component sizing, leading to a modified total energy production of 348,460 kWh/yr., indicating a decrease of 24.5%. Consequently, the proportion of energy consumed by the residential load decreases to 29.7%, compared to its initial representation of 29.34% of the total energy production. The operating cost experiences a reduction from \$17,497 to \$12,944, reflecting a 26.02% decrease. This reduction contributes to a decrease in both the NPC and the cost of energy (COE) from \$922,321 to

TABLE 5. Comparative Techno-economic summary for the different systems.

Parameters	Energy-Economics				
	Base Case	Qena City		Hurghada City	
		HRES without DSM	HRES with DSM	HRES without DSM	HRES with DSM
Generator (kW)	32	N/A	N/A	N/A	N/A
PV (kW)	N/A	58.1	52.6	49.5	30.2
Wind turbine (kW)	N/A	129	103	116	91
Batteries (kWh)	N/A	476	320	500	288
converter (kW)	N/A	24.9	19.8	29.2	19.7
Electrolyzer (kW)	N/A	30	30	3	9
HTank (kg)	N/A	140	140	120	260
Fuel Cell (kW)	N/A	60	40	80	40
COE (\$)	0.391	0.255	0.255	0.226	0.206
NPC(\$)	1.60M	1.04M	798,614	922,321	646,046
Operating Cost(\$/yr)	52,310	20,988	16,117	17,497	12,944
Renewable fraction(%)	0	100%	100%	100%	100%
Dispatch strategy	LF	LF	LF	CC	LF

TABLE 6. Total power production and consumption.

Components	Electricity Production				
	Base Case	Qena City		Hurghada City	
		HRES without DSM	HRES with DSM	HRES without DSM	HRES with DSM
Generator (kWh/yr)	135,407	N/A	N/A	N/A	N/A
PV(kWh/yr)	N/A	106,323	96,260	92,426	56,375
Wind Turbine(kWh/yr)	N/A	287,217	229,329	1,009	3,490
Fuel Cell(kWh/yr)	N/A	2,629	4,160	367,879	288,595
Total Production(kWh/yr)	135,407	396,169	329,749	461,314	348,460
AC Primary Load(kWh/yr)	135,254	135,216	103,655	135,146	103,679
Input Electrolyzer(kWh/yr)	N/A	29,072	44,218	10,382	39,459
Excess Electricity(kWh/yr)	153	214,785	168,646	302,398	192,826
Excess Electricity (%)	0.113	54.2	51.1	65.6	55.3
Unmet Electric Load(kWh/yr)	0	38.4	63.8	90.2	39.9
Unmet Electric Load%	0	0.0284	0.0615	0.0667	0.0385
Capacity Shortage(kWh/yr)	0	131	104	134	101
Capacity Shortage%	0	0.0971	0.1	0.0993	0.0976

\$646,046 and \$0.226 to \$0.206, respectively. This reduction represents 29.95% and 8.85%, respectively. The downsizing of components results in a reduction in total electricity production, subsequently minimizing excess electricity. After applying DSM, the surplus electricity decreases to 192,826 kWh/yr, down from the initial value of 302,398 kWh/yr. This adjustment means that the excess electricity now represents 55.3%, as opposed to the initial representation of 65.6% of the total energy production. Conversely, the energy consumed by the electrolyzer increases following the adoption of DSM, now representing 11.35% compared to the initial 2.3% of the total energy production. The suggested architecture has the capability to independently satisfy the load demand without

relying on external sources of energy generation, such as DGs, therefore the higher the value of the excess power, the better the system's dependability. TABLE 6 shows the details of unmet electric load and capacity shortage. The unmet load is described as the electrical load that the power plant is unable to support. This occurs when the supply is insufficient to meet the electrical demand. Furthermore, the difference between the power plant's actual operational capacity and the required operating capacity is known as a capacity shortage [77]. There were both unmet power demand and capacity limitations for the various cities evaluated in the different scenarios but the value of unmet load and capacity shortage is zero. These can be overcome by adding more storage

TABLE 7. Total Hydrogen production and consumption.

Quantities	Electricity Production				
	Base Case	Qena City		Hurghada City	
		HRES without DSM	HRES with DSM	HRES without DSM	HRES with DSM
Hydrogen Production(kg/yr)	N/A	626	953	224	850
Hydrogen consumption by FC(kg/yr)	N/A	552	874	212	733

units (such as batteries) to store the extra electricity produced during the day to meet demand for electricity during peak or irregular times of the day. To convert extra power into thermal energy, a resistive heater can also be integrated with the system [77]. TABLE 6 additionally shows the battery system's performance for various configurations at various locations in all scenarios. The amount of energy that passes through the storage system's throughput, or battery bank, in a year. After charging losses and before discharging losses, the change in the energy level of the storage bank is measured. The annual throughput for the various sites shows that Qena without DSM has the largest annual throughput while Hurghada with DSM the lowest. The costs of cycling the energy through the battery bank can be used to explain the battery wear cost. If the storage system's characteristics indicate that throughput is what limits storage life, HOMER's simulation will assume that the storage bank has to be replaced when its total throughput is equal to its lifetime throughput.

B. HYDROGEN PRODUCTION

TABLE 7 provides insights into the annual hydrogen production levels in different scenarios across various locations. Hydrogen production exhibits monthly variations at different sites, as outlined in TABLE 7, which illustrates both hydrogen production and consumption. In Qena city, the scenario with the highest hydrogen production, reaching 953 kg/yr post-DSM implementation, coincides with FC hydrogen consumption of 874 kg/yr., which is representing 91.7% of the produced hydrogen. This stands as the highest consumption rate among all scenarios. In contrast, the lowest production value is 626 kg/yr without DSM, with FC hydrogen consumption at 552 kg/yr, representing 88.2%. Similarly, in Hurghada city, the scenario adopting DSM records the highest hydrogen production and consumption values. The peak production is 850 kg/yr, with FC hydrogen consumption at 733 kg/yr, representing 86.2%. On the contrary, the lowest production value is 224 kg/yr without DSM, and FC hydrogen consumption is 212 kg/yr, constituting 94.6%. The hydrogen load effectively utilizes the surplus hydrogen produced across various sites. The results indicate that an increase in hydrogen production percentage correlates with higher fuel cell hydrogen consumption. This compensatory consumption addresses the shortfall in electricity production from alternative sources like PV and wind turbines, as detailed in TABLE 6, highlighting the electricity production from each source in all scenarios.

C. EMISSIONS

In the base case, CO_2 emissions reach the highest value at 124,654 kg/yr. However, the utilization of sustainable resources results in the complete absence of these emissions. Moreover, combining hydrogen with other RESs, such as PV modules and WT, significantly diminishes CO_2 emissions. In Qena city, both with and without Demand-Side Management (DSM), the system configuration leads to reductions in CO_2 emissions: 0.174 kg/yr. and 0.275 kg/yr., respectively.

In Hurghada city, the reduction values are 0.0666 kg/yr. with DSM and 0.230 kg/yr. without DSM. These results highlight the highest reduction values occurring in cases involving DSM.

X. CONCLUSION

This study provides a comprehensive analysis of the techno-economic and environmental implications of integrating a diesel generator (DG) with a hybrid renewable energy system (HRES), comprising photovoltaic (PV) modules, wind turbines (WT), and hydrogen fuel cells (FC). The analysis focuses on two prominent Egyptian cities, Qena and Hurghada, and explores the incorporation of demand-side management (DSM) to mitigate load, resulting in reductions in both net present cost (NPC) and cost of energy (COE). Using Homer Pro software, the study identifies the most cost-effective and optimal system configurations. Initially, the NPC for the diesel generator is the highest, standing at \$1.60 million. However, following the adoption of HRES, significant reductions are observed in both cities, with NPC values decreasing to \$1.04 million in Qena and \$0.922 million in Hurghada. Post-implementation hydrogen production levels are 626 kg/year in Qena and 224 kg/year in Hurghada. The application of DSM contributes to a substantial reduction in peak load demand, dropping from 28.31 kW to 21.71 kW, a decrease of 23.31%. In Qena, the NPC experiences an 11.3% reduction with DSM, further enhancing the economic efficiency of the system. The lowest NPC values in both Qena and Hurghada are achieved through the adoption of DSM, resulting in \$798,614 and \$646,046, respectively. In Qena, hydrogen production increases from 626 kg/year to 953 kg/year with DSM implementation, while in Hurghada, hydrogen production increases from 224 kg/year to 850 kg/year. Hydrogen production also serves to mitigate carbon dioxide emissions. Comparative analysis shows that the diesel system contributes 124,654 kg/year

of CO₂ emissions. In Qena, the WT/PV/hydrogen system with DSM reduces CO₂ emissions to 0.275 kg/year, and in Hurghada, emissions decrease to 0.0666 kg/year. The study highlights that the integration of HRES and DSM not only improves economic efficiency but also significantly reduces environmental impact.

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